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1st Check-in

I'm planning to build nanoGPT (by Andrej Karpathy) as my baseline and then layer on a few optimization tricks from Keller Jordan's modded-nanoGPT speedrun to see what actually helps, probably RoPE, RMSNorm, and maybe the Muon optimizer, though the exact list might change since I'm still working through the basics (refreshing my knowledge on nn, back-propagation, gradient descent). The rough claim I want to test is that some of these speedrun tricks genuinely improve the model at matched compute and some don't, so instead of just assuming the whole stack is justified I want to add them one at a time and see which ones actually result in improvements. For evaluation, I will mostly be looking at validation loss/perplexity across multiple seeds, keeping compute matched between conditions so it's a fair comparison, and I will do most of this on Colab at Shakespeare-character scale, maybe stepping up to a small GPT-2-scale run later if I need something more convincing, within a $20-50 budget. I don't have a strong background in ai/ml except from the only course I took with you. I will using this project as an opportunity to learn as much as I can on how llms actually work and hopefully reproduce my own scaled down version of GPT-2.

Research question: "Among RoPE, RMSNorm, and the Muon optimizer, which individually reduce validation loss relative to vanilla nanoGPT when trained at matched compute on Shakespeare-character data, and do any effects hold at small GPT-2 scale?"